Originalarbeit

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A Discrete Element Method Calibration Approach for Large Ore-flow Applications

Lukas Alexander Koch

Chair of Mining Engineering and Mineral Economics, Montanuniversität Leoben, Leoben, Austria

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Abstract: The major constraint for large discrete element method (DEM) simulations is the high demand for computation power. To keep the model in a feasible range of simulation time and to conduct the numerical calculations efficiently, the implemented material can be simplified and upscaled. After this, a calibration of the material is inevitable and sets the link between the simulation model and reality. Therefore, an alternative calibration approach is established in this contribution to mimic the flow behaviour of the original material with multi-spherical DEM particles. The calibration approach utilises flow zone dimensions, which were observed in block-caving operations. Additionally, a passive draw-point is used to calibrate the angle of repose. Applying the calibration approach for Kiruna iron ore showed that, with the help of the calibration approach, the material in the simulation could successfully reproduce the targeted flow zone and angle of repose.

Keywords: Discrete Element Method, Calibration, Cave mining, Ore-flow

Ein Kalibrierungs-Ansatz mit der Diskreten Elemente Methode für große Erzflussanwendungen

Zusammenfassung: Die größte Schwierigkeit bei großen Simulationen mit der Diskreten-Elemente-Methode (DEM) ist der hohe Bedarf an Rechenleistung. Um das Modell in einem machbaren Bereich der Simulationszeit zu halten und die numerischen Berechnungen effizient durchzuführen, kann das implementierte Material vereinfacht und hochskaliert werden. Danach ist eine Kalibrierung des Materials unumgänglich und stellt die Verbindung zwischen dem Simulationsmodell und der Realität her. Daher wird in diesem Beitrag ein alternativer Kalibrierungsansatz entwickelt, um das Fließverhalten des Originalmaterials mit multikugelförmigen DEM-Partikeln zu imitieren. Der Kalibrierungsansatz verwendet Fließzonen-Dimensionen, die im Blockbruchbau-Verfahren beobachtet wurden. Zusätzlich wird ein passiver Abzugspunkt zur Kalibrierung des Schüttwinkels verwendet. Die Anwendung des Kalibrierungsansatzes für Kiruna-Eisenerz zeigte, dass das Material in der Simulation mit Hilfe des Kalibrierungsansatzes die angestrebte Fließzone und den Schüttwinkel erfolgreich reproduzieren konnte.

Schlüsselwörter: Diskrete-Elemente-Methode, Kalibrierung, Bruchbau, Erzfluss

1. Introduction

There are several challenges when investigating large models containing millions of particles with the discrete element method (DEM). One of the most important challenges is the high demand for computing capacity. First of all, software with a large parallelisation capability is thus necessary, preferably with the possibility to run the simulation on a graphics card (GPU). In this contribution the software EDEM by Altair [1] was used. This contribution further concentrates on two points: simplifying the material in the model to reduce the high demand for computing power and subsequently calibrating the simplified material to yield realistic results as if it was the original material.

A DEM calibration can compensate to some extent for simplification measures undertaken beforehand. The calibration sometimes represents the largest part and a major difficulty in large-scale DEM simulations [2, 3]. For a DEM simulation, the material macro properties are distinguished from the DEM micro parameters [4]. The macro properties describe the macroscopic behaviour of the bulk material, e.g. the angle of repose or the flow zone dimensions in a cave mine. The DEM micro parameters, on the other hand, are the input parameters for the software which lead

Dipl. Ing. L. A. Koch (✉)
Chair of Mining Engineering and Mineral Economics,
Montanuniversität Leoben,
Erzherzog Johann-Straße 3,
8700 Leoben, Austria
lukas.koch@unileoben.ac.at

to certain macro properties. In the bulk calibration process, the micro parameters are calibrated in a sensitivity study to achieve the desired macro behaviour of the material. As the macro behaviour is usually dependent on more than one parameter, it is recommended to conduct more than one calibration experiment, each being representative for a respective application [5–7]. The combined results lead to the unique parameter set for the material of interest.

1.1 Requirements

An issue for material calibration in large models like cave mines is that there is no standardised calibration routine, and there are not many DEM models with a comparably large size available in the literature. Some authors [8–15], however, have approached the topic of ore-flow in cave mines via DEM. Some sources talk about non-calibrated or non-validated simulations [8, 9] or do not mention the underlying calibration method [10]. Yet, the basis for the calibration of the models was mostly small-scale physical tests. There are, however, issues with upscaling the results from small-scale physical models [16]. As the particle sizes in cave mines range from fines to large boulders, simple small-scale angle of repose tests are also difficult or even not feasible to conduct. For these reasons, a DEM calibration method based on full-scale mine data is needed for this kind of large model. This was approached by very

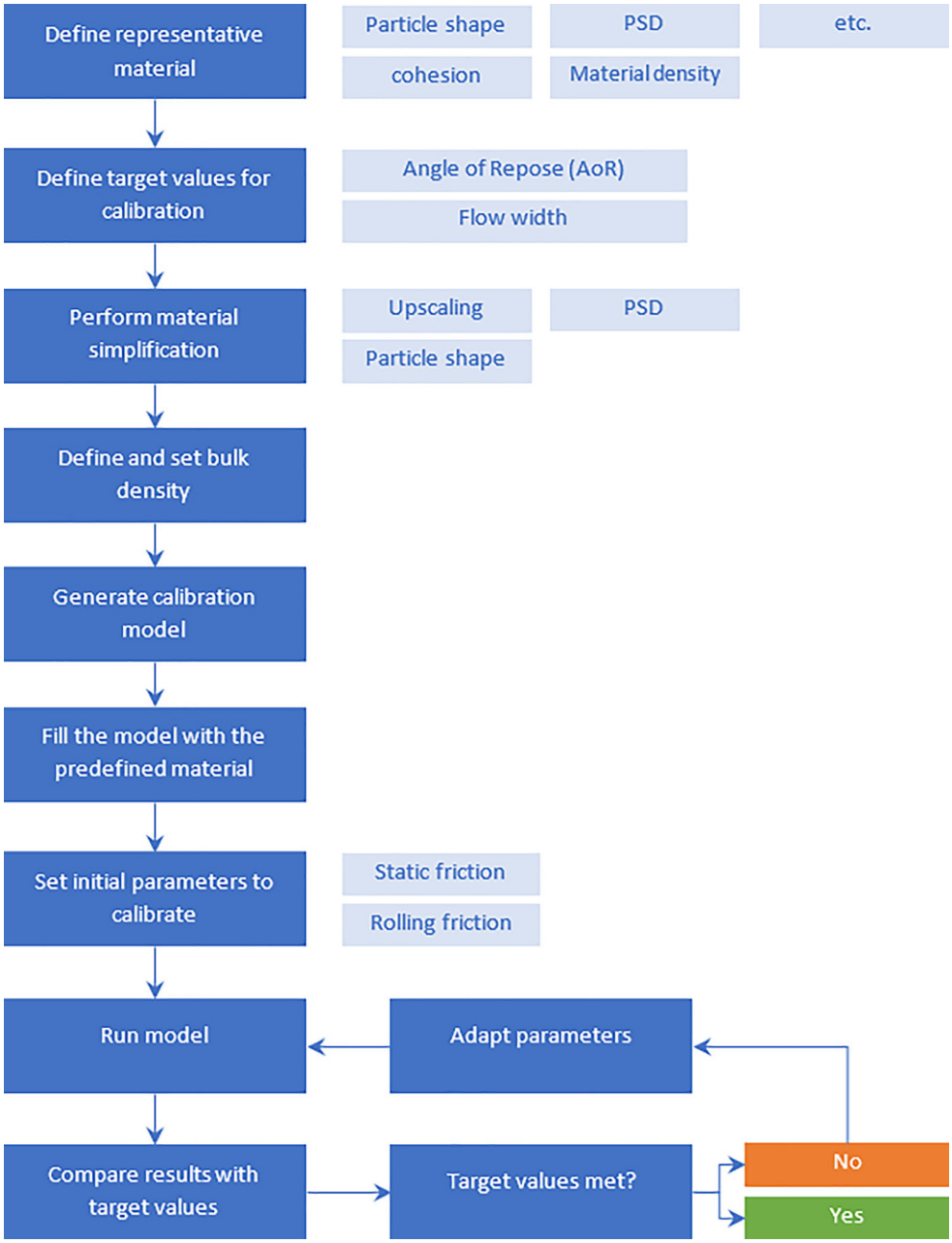


Fig. 1: Flow chart of the calibration process

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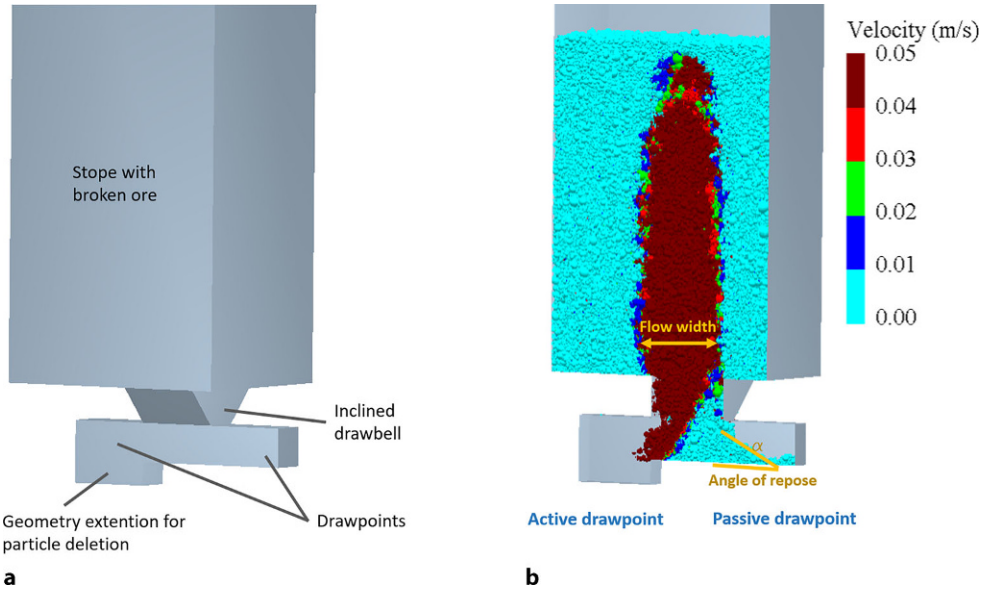


Fig. 2: Calibration model: **a** the CAD model, **b** the cross-cut of the running model. The flow width is obtained from the active draw-point and the angle of repose from the passive draw-point

few authors, like DeGagné and McKinnon [15], who calibrated the material by adapting the friction values until the material behaviour matched the draw responses seen in sublevel caving (SLC) rings. For free flow situations like in a large stope with dimensions of several tens of meters, like in block caving (BC), reliable DEM calibration approaches have not been found yet. Establishing a reliable calibration approach is the aim in this contribution.

1.2 Approach

The idea of the calibration approach is to generate a virtual material that possesses the same flow behaviour as the real physical one. Therefore, the physical one as well as its known macroscopic features need to be determined in the beginning. To decrease the computational demand and to enable a suitable DEM simulation, the virtual material usually needs to be simplified by different measures. The contact parameters, which are mainly the friction values (referring to a common Hertz-Mindling contact model), are then adapted in the calibration process, and the calibration simulation is repeated until the macroscopic behaviour corresponds to the behaviour of the physical material. When the virtual material behaviour fits, the calibrated parameter set is found, and the calibration is finished. If simplification measures are used too extensively, a successful calibration may not be possible.

Figure 1 summarises the systematic of this contribution's calibration approach. First, the basic properties of the original material that is expected to be encountered in the mine need to be defined (particle shape, particle size distribution, density etc.). Then the target values for the calibration are determined, which are the angle of repose and the draw-zone width, as explained later in more detail. To keep the number of required calibration simulations as small as possible, both calibration values are implemented in a single calibration model. The material is then simplified to enable the calculation of the model. Subsequently, the simplified material is inserted with a predefined bulk density into the calibration model. A sensitivity study is run by iteratively adapting the friction parameters and running the simulation until the target values of the calibration are met. To keep the number of simulations low, both chosen target values are evaluated in a single model. If the results correspond to the target values, the calibration is finished.

2. Calibration Approach

The presented calibration model is based on experiences on movement zones above a draw-point, made by Laubscher [17] in BC operations. Further on, the angle of repose at the draw-point is compared to a representative one found in the mine. The model, seen in Fig. 2a, consists of a stope of 25 × 30 × 40 m with broken material. On the bottom of the model, there is a draw-bell with a 45° inclination and two attached draw-points. The vertical distance from the floor of the extraction point to the bottom of the stope above is 10 m. Material is extracted from one—active—draw-point, which evokes a movement zone in the material pile above. From the second—passive—draw-point material flows out and stabilises at a certain angle of repose. The draw-points have a quadratic cross-section of 5 × 5 m and a straight roof. These dimensions were taken from typical BC layouts [18, 19].

To extract material from the active draw-point, the material is removed after the outflow through a vertical extension below the draw-point. Consequently, the overlying material begins to flow towards the draw-point, and a movement zone evolves. Material is allowed to flow out until the movement zone breaks through the top of the model, which is filled to a height of 40 m (see Fig. 2b). After this breakthrough point, the zone above the moving particles is destabilised. The simulations were then analysed and compared. Depending on the parameter set, the

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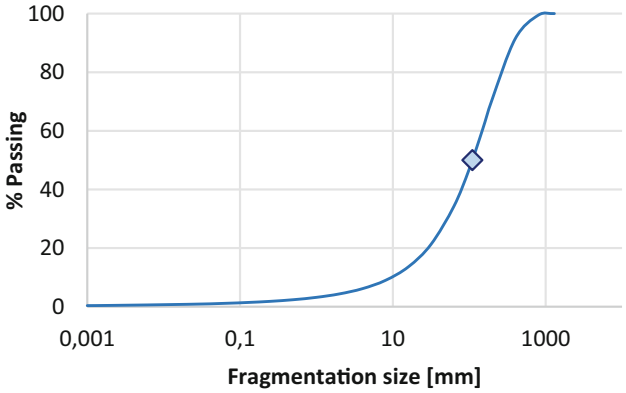


Fig. 3: Fragmentation curve of Kiruna SLC operation according to Wimmer et al. [21]

breakthrough was reached at different extraction states, but generally approximately around 4 kt of extracted material. The width of the movement zone was analysed by measuring the flow width of the active area within the block. The advantage of this model is that the angle of repose is additionally obtained at the passive draw-point.

2.1 Determination of a Representative Material

The first step in replicating material in a DEM model is to determine the representative physical material in nature. Therefore, the fragmentation size and the particle shape are estimated. The presented approach is utilised in ore-flow investigations for the Kiruna mine, researching a novel mining method called Raise-Caving [20]. The particle size distribution (PSD) is taken from available literature sources [21] stating the Kiruna SLC situation with an x_{50} value of 110 mm (see Fig. 3). Future investigations should also target other fragmentation groups, which include coarse and homogeneous, coarse and inhomogeneous groups, and the influence of water and cohesion.

Regarding the particle shape, pictures from the material at SLC draw-points in the Kiruna mine (see Fig. 4) gave the impression of a rather angular material shape. This information was seen to be sufficient for modeling, as the overall material behaviour calibration compensates for a certain lack of particle-shape detail.



Fig. 4: Fragmentation of a standard draw-point in Kiruna

2.2 Target Values for Calibration

The basis for the calibration is the experience made in many BC operations by Laubscher [17]. Laubscher showed that the size of extraction zones above a draw-point depended on the PSD and the dimensions of the draw-point (see Fig. 5). In this table, the loading width indicates the width of the draw-point. The rock mass class defines the size of the caved rock mass. Wide flow zones above the draw-points are achieved when the material is very coarse. The flow zone gets slimmer with finer material.

The fragmentation data seen in Fig. 3 was combined with a draw-point width of 5 m, which resulted in a draw-zone diameter of approximately 10 m. This was consequently set as the target value for the calibration. For the angle of repose, Määttä et al. [22] showed for 3D-laser-scanned Kiruna draw-points that a normal and stable gravity flow took place when the angle of repose varied between 30–35°. Beakawi [23] analysed that, for coarse gravel material, values of 25–45° could be possible, depending on the content of fines. It was seen in literature that the angle of repose increases for finer material [23, 24]. As a combination of this information, a range of 33–37° was set as a target for the angle of repose of the investigated fine material group.

2.3 Simplification Measures on Particle Properties

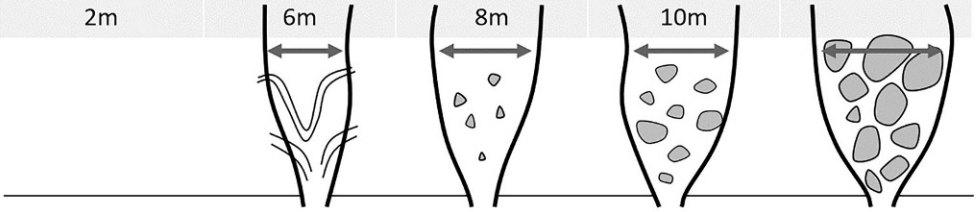
2.3.1 PSD and Particle Shape Simplification

The simplification measures are applied to the determined material. These need to be implemented within their limits to yield realistic results. The first simplification measures are to neglect fines and simplify the particle shape. An issue with implementing fines is the excessive calculation demand as these contain billions of relatively small (and, respectively, computationally intensive) particles. In state-of-the-art simulations, fines are commonly neglected. The lack of this part of fragmentation needs to be compensated with a proper calibration. Additionally, it was seen in the course of this contribution that the implementation of large boulders—although reducing the number of particles in a simulation—affected the computational speed negatively, as the time step was reduced to maintain the particle overlaps in a relatively small range. Boulders in Kiruna are defined as material blocks larger than 0.7 m. However, the fraction of particles larger than 0.5 m is already very small and was therefore neglected in the basic models. The implemented PSD consequently followed the curve of Fig. 3 by only considering the sizes 50–500 mm. To generate this PSD in the DEM model, four different particle size groups were used—50, 150, 300, and 500 mm.

The particle shape has a significant effect on the bulk material behaviour; however, literature references [3, 4] show that an exact replication of the particle shape in DEM simulation is often not necessary. Complex particles can be replicated in DEM simulations by simplified particle shapes. For example, multiple spheres can be consolidated into multi-sphere particles. They are inseparably fixed together and

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Rock Size Range	0.01-0.3m	0.1-2m	0.4-5m	1.5-9m
Loading Width	Isolated Drawzone Diameter			
5m			11.5m	13m
4m		9m	11m	12.5m
3m	6.5m	8.5m	10.5m	12m
2m	6m	8m	10m	



Loading Width	Maximum/Minimum Spacing of Drawzones			
5m				24/14m
4m		15/8m	20/11m	22/13m
3m	10/5m	13/7m	18/10m	21/12m
2m	9/4m	12/6m	16/9m	

Fig. 5: Isolated draw-zone diameter (flow width) for different size ranges of caved rock mass at certain loading widths after Laubscher [17, 18]

consequently represent a single particle. It could be shown that the flow behaviour of most common non-spherical particles (excluding highly complex shapes, such as fibers, plates, interlocking particles etc.) can be reproduced with this approach when calibrated properly [4] (e.g. flow from a filled container). It is stated that a particle clump consisting of two spheres is already much more precise than a simple single sphere. Using more than four spheres does not increase the accuracy significantly but increases the computational effort. A compromise between realistic and efficient simulations can be made by implementing a manually created particle mix with different numbers of spheres per particle. In this contribution, a manual material mixture was chosen similar to the one of Coetzee [4], who found that this can reproduce a good accuracy for angular material. As a compromise between calculation efficiency and accuracy, seven different particle shapes with one to four spheres were implemented (see Fig. 6).

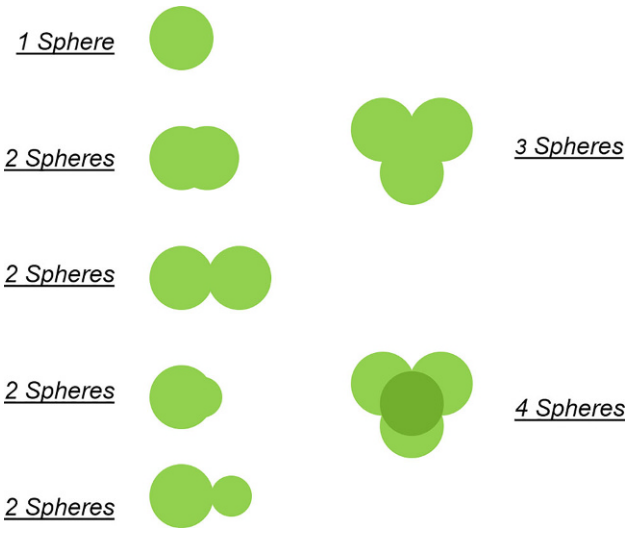


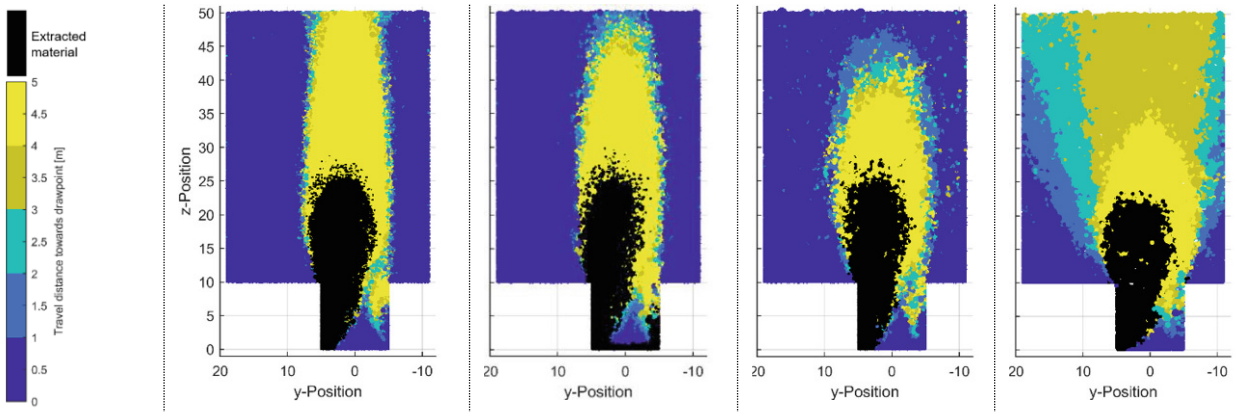
Fig. 6: Particle shapes containing one to four spheres for the RC models

2.3.2 Shear Modulus, Coefficient of Restitution, and Upscaling Factor

A further point to speed up the simulation is to reduce the shear modulus in the model compared to reality. This can yield massively faster results. If applied within its limits, it does not change the quality of the result for many applications [25, 26]. The shear modulus reduction was tested by Brandner and Koch [27]. As a consequence, the shear modulus was reduced in this contribution from originally 34 to 5 GPa. Considering the value of the coefficient of restitution (COR), it was found by different authors [26, 28] that DEM simulations showed an insensitivity to the variation of the COR. Its exact determination can additionally only be done individually for every particle with drop tests. This was seen to be impractical for a calibration process, which is why literature research on this parameter was done. Barrios et al. [29] investigated similar material properties to crude iron ore and used the same software as in this contribution (EDEM). Therefore, the found COR value of 0.51 served as the guiding value, with a value of 0.5 implemented in this work.

The calibration model with 25 × 30 × 40 m dimensions still contains over a hundred million of the simplified particles, which is not feasible to compute with contemporary workstations. A solution to reduce the number of particles is to apply particle upscaling. The idea is to replace the particles by larger ones. Up to a certain point, specifically depending on the application, this was seen not to affect the results adversely, especially when a calibration is conducted subsequently. For large stope applications, this was tested by Brandner and Koch [27] and seen to

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Upscaling factor	*2*	*3*	*4*	*6*
PSD	*100-1,000mm*	*150-1,500mm*	*200-2,000mm*	*300-3,000mm*
Speed-up ratio	*1*	*5.3*	*16.4*	*33.8*

Fig. 7: The original material of 50–500bmm is upscaled in size by factors from 2 to 6. Black particles indicate the extracted zone while the colours show the movement zone. It shows that too large upscaling factors change the simulation results drastically

be reasonable up to an upscaling factor of 3. This value means that, instead of 50–500 mm, the implemented particles have sizes of 150–1500 mm. However, the PSD in this study differs to the present one, which is why it is tested again with the present PSD.

As a comparison value, the centrally located black extraction zone and the movement zone on top of it were used, as seen in Fig. 7. This area is focused with minimal regard for minor diverging boundary effects. In order to achieve a speed-up effect, the method introduces some idealisations. The potential slight distortions introduced by the upscaling approach need to be acknowledged and accounted for in the analyses. This awareness allows us to place greater emphasis on trends and relative comparisons rather than relying solely on the absolute results. With that context, Fig. 7 shows that the upscaling factors of 2 and 3 deliver similar results for the flow zones and the upscaling factors of 4 and 6 lead to a more distorted flow image than the previous ones. This leads to the conclusion that an upscaling factor of 3 is the limit for the present application. As it was 5.3 times faster than factor 2, an upscaling factor of 3 was consequently used in this contribution (being aware of its influences).

2.4 Parameter Study for a Realistic Bulk Density

Bierwisch et al. [30] found that it is mandatory for a DEM model for the bulk density to be the same as the original one. Otherwise the granular behaviour might deviate due to a changed amount of potential energy in the system. There are various uncertainties related to the bulk density of the broken ore in the stope due to blasting and compaction mechanisms. It is, however, reasonable to assume that the bulking factor K of broken ore in the region of the draw point will be in the range of $1.3 < K < 1.5$. The bulk density of the broken ore will therefore fall in the range of 3600–3130 kg/m^3.

To set up the simulation model, the material is dropped into the stope from the top. It was seen that the predefined friction values during the filling process already influenced the bulk density of the model. Higher friction led to a lower bulk density and vice versa. To find adequate friction values for the filling process, a parameter study was done. A box was filled several times with different friction values. The static and rolling friction values were varied during each filling process, from 0.1, to 0.5, and 0.9. This is done for both friction values, which leads to a full set of nine parameter combinations (see Fig. 8; also being aware that commonly the rolling friction should be set lower than the static friction value). The target bulking density of 3600–3130 kg/m^3 can be achieved by four different parameter combinations. To compensate for any larger voids across the entire stope, the lower value of the range was chosen. Therefore the used density was 3111 kg/m^3, achieved by the parameter set of 0.5 (static friction) and 0.1 (rolling friction).

2.5 Results of the Actual Calibration Model

The target values for the calibration are 10 m flow width and 33–37° angle of repose. To find a set of friction pa-

Property	Value								
static friction	0.1	0.1	0.1	0.5	0.5	0.5	0.9	0.9	0.9
rolling friction	0.1	0.5	0.9	0.1	0.5	0.9	0.1	0.5	0.9
Bulk density [kg/m^3]	3,418	3,312	3,352	3,111	2,840	2,818	2,993	2,641	2,620

Fig. 8: Parameter study—effect of friction coefficients during the filling process on the bulk density

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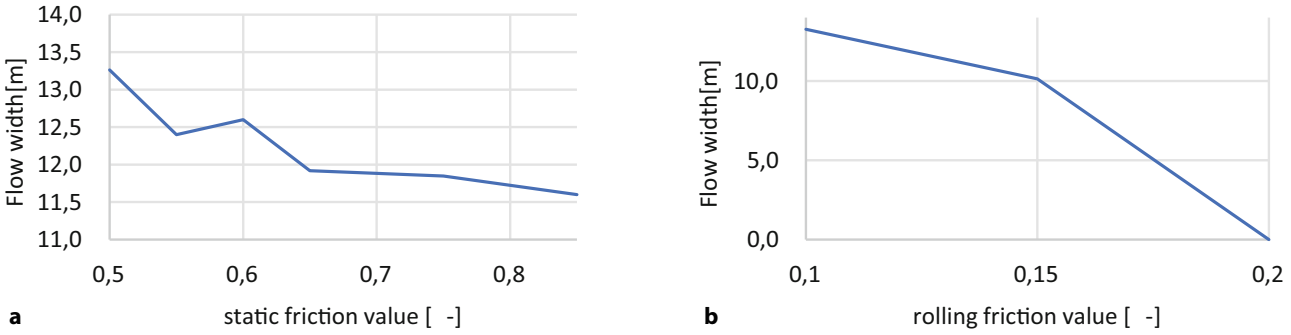


Fig. 9: Correlation of the flow width with **a** the static friction value at a constant rolling friction of 0.1 and **b** the rolling friction value at a constant static friction of 0.5

Fig. 10: Comparison of the best-fitting data sets of the conducted calibration models

Static friction [1]	0.60	0.65	0.70	0.75	0.75	0.85
Rolling friction [1]	0.12	0.10	0.12	0.10	0.12	0.10
Angle of repose [°]	35.5	34.9	35.6	35.7	35.8	34.4
Flow width [m]	11.4	11.9	11.9	11.8	10.9	11.6

rameters that produce this material behaviour, the effect of static and rolling friction parameters was investigated individually. One parameter was locked at the time, and the other one was varied to see the effect on the bulk behaviour. First, the rolling friction was kept at a value of 0.1 with the static friction increased from 0.5 to 0.85 in steps of 0.05. The result in Fig. 9a shows that the flow width decreases with growing static friction, as expected. The targeted flow width of 10 m is, however, difficult to achieve by only adapting the static friction.

As a second step, the rolling friction was varied from 0.1 to 0.15 and 0.2 with the static friction kept at a value of 0.5. Figure 9b shows that a friction value of 0.15 results in a flow width of 10.1 m, but at an angle of repose 43°.

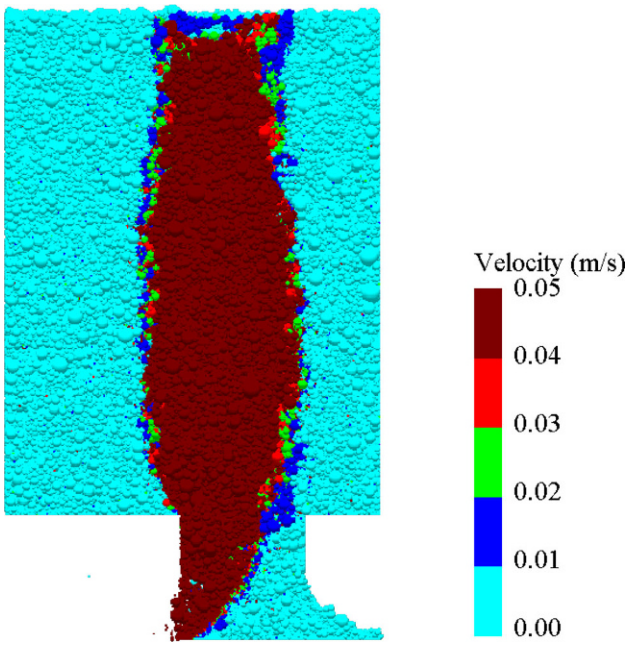


Fig. 11: Cross-section of the calibration model and the flow zone for a static friction of 0.75 and a rolling friction of 0.12

A rolling friction of 0.2 even led to immediate hang-ups at the draw-point. This concludes that increasing the rolling friction reduces the flow width, but a too-high value provokes an unnatural behaviour.

Further simulations were conducted with different parameter sets, varying both friction parameters. The calibration results can be seen in Fig. 10, with the parameter set 0.75 static friction and 0.12 rolling friction approximating the target values in a suitable way. This set resulted in an angle of repose of 35.8° and a flow width of 10.9 m (see Fig. 11). Consequently, this friction parameter set is used in subsequent models for a detailed investigation of key issues related to the ore-flow. The calibration has been applied by Koch et al. [20] to investigate the draw-bell inclination and the draw-point spacing on the ore-flow. A summarising table of all used input parameters and the calibrated values can be seen in Table 1.

3. Conclusion, Summary, and Outlook

The contribution showed a successful way to calibrate material in large discrete element method (DEM) models for cave mining. This was based on full-scale mine data, which included the flow width within a stope and the angle of repose of a passive draw-point. The model consists of a large stope with broken material, an underlying draw-bell with a 45° inclination, and two draw-points at the bottom. One draw-point is actively worked and consequently forms a flow zone of broken material in the draw-bell above it. The other draw-point is passive, and the outflowing material forms an angle of repose.

The calibration targeted the flow behaviour of the iron ore in the Kiruna mine. Therefore characteristic properties concerning the particle shape and particle size distribution were implemented in the software. Simplification measures were applied within the tested limits to facilitate the computation. A parameter study was conducted to find suitable friction parameters for the filling process in order

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TABLE 1
Summary of parameters for the DEM calibration model

Property	Value
Material solids density	4700 kg/m^3
Resulting material bulk density	3156 kg/m^3
Static friction for filling process	0.5
Rolling friction for filling process	0.1
Particle shape	Multishperical; varying in 7 different constellations with 1 to 4 spheres; see Fig. 6
Original PSD	According to [21] (x_{50} of 110 mm); see Fig. 3
Simplified PSD	50–500 mm (50, 150, 300, and 500 mm)
Used upscaling factor	3
Target flow width	10 m
Target angle of repose	33–37°
Static friction result from calibration	0.75
Rolling friction result from calibration	0.12

to set up the model for later discharge modeling representing a realistic bulk density. A value of 3159 kg/m^3 could be achieved with a parameter set of 0.5 and 0.1 for static and rolling friction, respectively. These friction parameters, however, were not the ones set for the subsequent material discharge simulations and were only used to perform the filling process as part of the model preparation. For the actual calibration, the implemented particle size distribution was combined with draw-zone width experience from block caving. This led to the target value for calibration of 10 m draw-zone width. Further, the angle of repose was calibrated based on typical Kiruna draw-point angles and values from literature of 33–37°. The calibration could be successfully achieved with a suitable parameter set of 0.75 static friction and 0.12 rolling friction. These calibration results should further be validated at a future stage to control their quality, which can be done by representative tests in the mine and by evaluation of experts in the field of DEM and ore-flow. It was also seen that the bulk density plays a major role as it affects the results, thus leading to an open point for future research regarding a more detailed determination of the physical bulk density inside block caving stopes.

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